YG - Management of Massive Haemorrhage in Adults (not obstetrics)

ONGOING MASSIVE HAEMORRHAGE WITH CIRCULATORY COMPROMISE
e.g. Systolic BP <70mmHg; Ongoing bleeding >150ml/min; Massive GI Bleed; Massive haemorrhage in Major Trauma

Dial 2222

"Massive Haemorrhage" and your location

GET CONSULTANT HELP

STOP THE BLEEDING

Get a surgeon

Bleeding Control - Immediate

- Direct pressure
- Elevate bleeding limb
- Tourniquet
- Splint fractures: Binder/KTD
- Haemostatic dressings e.g. Celox
- Sengstaken tube
- Permissive hypotension
  - *Consultant decision only*
  - *Maximum time limit 90mins from bleeding **onset***

Support Coagulation

Tranexamic Acid
- 1g iv over 5 mins
- Then another 1g over 8hr

Calcium Chloride 10%
- 10ml after 4th unit blood

Warfarin?
- Beriplex (PCC) 25-50u/kg*
- Vit K 5mg iv

Clopidogrel?
- Platelets 2 units*

Dabigatran?
- Praxbind*

Direct Xa inhibitors?
- PCC 25-50u/kg*

Variceal Bleed?
- Terlipressin 2mg iv

Bleeding Control - Definitive

- Surgery
- Endoscopy
- Interventional Radiology

RESUSCITATE

Airway; Breathing; Circulation

Full Monitoring

In ED, move to resus

ICU Involvement

GIVE WARMED BLOOD

Give 2 units Emergency Group O

Consider likely need for further blood products now

Lab will need time to defrost FFP/arrange crossmatch etc.

SPEAK TO LAB URGENTLY

Tell them what you need

Ext 5880 (in hours)

Page 043 (out of hours)

SEND SAMPLES TO LAB

Crossmatch:

2 samples from 2 sites in 2 bags

FBC, Coag screen, U+E, Ca^{2+} ABG

4:4:1 +/- ROTEM Guided Therapy

4:4:1 of PRC:FFP:Platelets

If Fibrinogen <1.5 give 2 bags cryoprecipitate

ROTEM

Order additional units as follows:

Caution using ROTEM in patients taking antiplatelet agents

FIBTEM A5 <10mm	CRYOPRECIPITATE 2 Bags
EXTEM A5 - FIBTEM A5 <30mm	PLATELETS 1 Bag

After administering initial FFP in 1:1 ratio +/ - cryoprecipitate:

EXTEM A5 >40mm and EXTEM CT >80secs	FFP Give 4 additional units

REASSESS after every 4th unit

Repeat all bloods

Order more products if needed

Warm the patient

- Warm all iv's
  - Ideally Belmont
  - At least Ranger
- Use Bair Hugger
- Target 36^0C

Blood Tests – aim:

- Check recent results

Hb	80-100g/L
pH	> 7.35
Ca^{2+}	>1 mmol/L
Platelets	>75 x 10^9/L
PT	< 18sec
APPT	< 45sec
EXTEM CT	**<80sec**
EXTEM A5	**≥40mm**
FIBTEM A5	**>10mm**

If Fibrinogen <1.5 give 2 bags cryoprecipitate

WATCH Ca^{++} WATCH K^+

WHERE DO I GET BLOOD FROM?

2 Units 'O' will be issued and brought to location by:

Nurse (Ward) ODO (Theatre/Cybi) Porter(ED)

- **Further blood and products by liaising with lab:**
  - Give crossmatched if available (may take time)
  - Give Group Specific if no crossmatched (quicker)
  - Otherwise Give Emergency O (quickest)
- **Coagulation products marked with * (above) need discussion with haematologist (via switch on ext 100)**

Consider Cell Salvage

1 unit PRC = 250ml salvaged blood

STAND DOWN

Once bleeding stopped and patient stable

- Inform lab
- Return unused components
- Complete documentation for all blood products
  - (including audit proforma)
- Thromboprophylaxis when patient stable